# Valuing global wildfire $PM_{2.5}$ mortality caused by climate change

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**Abstract**

The social cost of carbon dioxide (SC-$CO_2$) estimates the monetized damages from an additional ton of $CO_2$ emissions. Existing global damage estimates capture the effect of climate change on temperature-related mortality, agriculture, energy, and sea-level rise, but exclude the impact on wildfires. We address this gap by estimating the mortality damages attributable to fine particulate matter ($PM_{2.5}$) air pollution from climate-induced wildfires. To do so, we harmonize output from Earth System Models (ESMs) to derive country-specific relationships between global mean temperature (GMT) change and population-weighted $PM_{2.5}$ exposure, and we parameterize a wildfire smoke mortality damage sector within the GIVE integrated assessment model. In Monte Carlo simulations that sample key damage-function parameters, we estimate a mean SC-$CO_2$ from wildfire smoke mortality of \$50/t$CO_2$ (95% conditional interval: \$18.5–\$102.6). This raises the mean global total SC-$CO_2$ by 24%, from \$208 to \$258 per ton $CO_2$ at a 2% discount rate. The regions contributing most to the global wildfire $PM_{2.5}$ damage total are Sub-Saharan Africa, the Former Soviet Union, and the United States, driven by large estimated increases in population-weighted $PM_{2.5}$ exposure per degree of warming. Since we account for only the excess mortality due to wildfire smoke, our estimates provide a lower bound of the SC-$CO_2$ attributable to climate-induced wildfires.

Long-term exposure to fine particulate matter air pollution ($PM_{2.5}$) is associated with increased all-cause mortality (Orellano et al., 2024; Pope et al., 2020). Wildfire smoke is a significant and growing source of $PM_{2.5}$ (Park et al., 2024), and climate change is predicted to increase wildfires, smoke exposure, and associated mortality in many regions (Pierce et al., 2017; Zhao et al., 2025). However, the mortality damages from climate-induced wildfire smoke are missing from existing global estimates of the social cost of carbon dioxide (SC-$CO_2$). The Greenhouse Gas Impact Valuation Estimator (GIVE) integrated assessment model estimates the SC-$CO_2$ with four damage sectors—temperature-related mortality, agriculture, energy, and sea-level rise (Rennert et al., 2022)—but it ignores climate-induced wildfire smoke. Adding an additional damage sector to GIVE, Qiu et al. (2026) estimates a partial SC-$CO_2$ accounting for wildfire smoke mortality in the United States, finding it to be roughly half of the *total* damage. No global estimate yet exists.

This study develops a global damage function relating global mean temperature (GMT) change to wildfire smoke mortality, which we integrate into GIVE as a fifth damage sector. To parameterize this function, we estimate country-level relationships between GMT change and population-weighted $PM_{2.5}$ exposure using data from five global gridded analyses of Earth Systems Models (ESMs).[1] Process-based ESMs explicitly represent vegetation growth, fuel loads, fuel combustibility, wildfires, and chemical transport. In contrast, Qiu et al. (2026, p. 5) apply two separate empirical models that link temperature to wildfire activity and wildfires to $PM_{2.5}$ concentrations, so their estimates do not account for the effects of past wildfires on future wildfire severity, intensity, or vegetation composition and species shifts that may substantially influence projected wildfire emissions, smoke, and associated mortality. The main advantage of our approach is that ESMs explicitly model those feedback dynamics on a global scale. The main disadvantage is that ESMs are computationally intensive, so global, gridded land-atmosphere-wildfire output is limited. We estimate the the relationship between GMT and wildfire smoke exposure across 43 historical and future simulations. We combine these estimates with a concentration-response coefficient to compute mortality damages and a global SC-$CO_2$ (Section 3). Our analysis of ESM output makes our global damage function a useful complement to Qiu et al. (2026)'s empirical approach, and we estimate a similar partial SC-$CO_2$ attributable to wildfire smoke mortality in the US.

Following Rennert et al. (2022), we assess uncertainty in our SC-$CO_2$ estimates via Monte Carlo simulation, sampling over both the concentration-response coefficient and the country-level GMT-$PM_{2.5}$ coefficients (Section 3.4). We discuss the strengths and limitations of our approach relative to Qiu et al. (2026) (Section 8) and compare our U.S. partial SC-$CO_2$ estimate to theirs (Section 9). In sensitivity tests, we also report our results excluding data from

[1] Pierce et al. (2017) use CESM, Park et al. (2024) use CLASSIC, SSiB4, and JULES, and Zhao et al. (2025) use CESM2, CESM2-WACCM, EC-Earth3-CC, and EC-Earth3Veg.

individual ESM models (Section 10).

# 1 Results

We present results in two parts. First, we present GMT-$PM_{2.5}$ relationships at the *grid-cell level*. These estimates of the change in the concentrations of wildfire $PM_{2.5}$ ($\mu$g / $m^3$/ year) per 1°C increase in GMT reveal where process-based models predict that warming most strongly increases the exposure of wildfire PM $_{2.5}$ (Figure 1). Second, we report the global SC-$CO_2$ attributable to wildfire $PM_{2.5}$ mortality using our GIVE wildfire smoke damage function (Figure 2). This function uses our estimate of *country-level* GMT-$PM_{2.5}$ relationships describing the change in population-weighted per-capita wildfire $PM_{2.5}$ exposure ($\mu$g/$m^3$/yr/person) per 1°C increase in GMT (Figure 3; Section 3.1).

In Figure 1, the strongest GMT-$PM_{2.5}$ relationships are in boreal North America, boreal Asia, the Congo Basin, and the Amazon basin—regions[2] where climate change has been shown to increase wildfire activity through rising temperatures and drying fuel loads (Park et al., 2024; Wimberly et al., 2024; Zhao et al., 2025). Moderate positive relationships are seen across temperate North America, Northern Africa, Europe, and parts of South and Southeast Asia. A minority of grid cells show near-zero or negative relationships in parts of Africa, South America, and Australia. In sub-Saharan Africa, there is evidence that competing mechanisms[3] may offset one another as well as disagreement among ESMs on the impact of climate change in this region (Park et al., 2024). Parts of Australia shows unexpectedly flat or negative relationships, driven by flat or declining predicted wildfire $PM_{2.5}$ in the data from Zhao et al. (2025) and Pierce et al. (2017) (Figure 5).

Figure 2 reports the global SC-$CO_2$ attributable to wildfire $PM_{2.5}$ mortality, with comparison to contributions from existing GIVE damage sectors. Adding to the uncertainty already captured in GIVE, we incorporate conditional uncertainty in the wildfire $PM_{2.5}$ damage function parameters (see details in Section 3.4). The wildfire $PM_{2.5}$ SC-$CO_2$ has a mean of \$50/ton $CO_2$ (95% conditional interval: \$18.5–\$102.6) at a 2% discount rate (Panel A). Panel B shows that wildfire $PM_{2.5}$ mortality represents the third-largest damage sector after temperature-related mortality (\$96) and agriculture (\$100), increasing the total global SC-$CO_2$ by 24%, from \$208 to \$258. Our U.S. partial SC-$CO_2$ from wildfire $PM_{2.5}$ is somewhat smaller to that found by Qiu et al. (2026) (Section 9). Our decomposition reveals substantial regional heterogeneity in

[2]Climate change is expected to increase wildfire activity in large parts of Russia and Canada (high-latitude boreal forests areas) driven by warming and increasing drought (Zhao et al., 2025). Wildfire activity in the Congo Basin tropical forests has also increased, with year-to-year variation synchronized with rising temperatures and vapor pressure deficit, alongside deforestation (Wimberly et al., 2024).

[3]Higher temperatures may increase ignition potential while also driving evapotranspiration that may reduce grass productivity and fuel loads (Park et al., 2024).

$SC\text{-}CO_2$ estimates (Table 1).

The $SC\text{-}CO_2$ results are driven by the *marginal* excess death associated with wildfire smoke, computed by subtracting projected excess death in a baseline model from the same projection with one additional ton of $CO_2$, where excess death is a linear function of GMT change (Equation (2)). Accordingly, warming induces an average of 289,295 excess deaths per year globally at 1.5°C, rising to 696,794 at 3°C and 948,800 at 4°C. We compare our U.S. excess death estimates and damage function shape to Qiu et al. (2026) in Section 9. Relative to the Qiu et al. (2026) quadratic function, our linear function projects more deaths at low warming levels and fewer deaths at high warming levels. However, on average, the Qiu et al. (2026) quadratic yields higher *marginal* excess deaths across most of the observed temperature range, which drives the $SC\text{-}CO_2$ results.

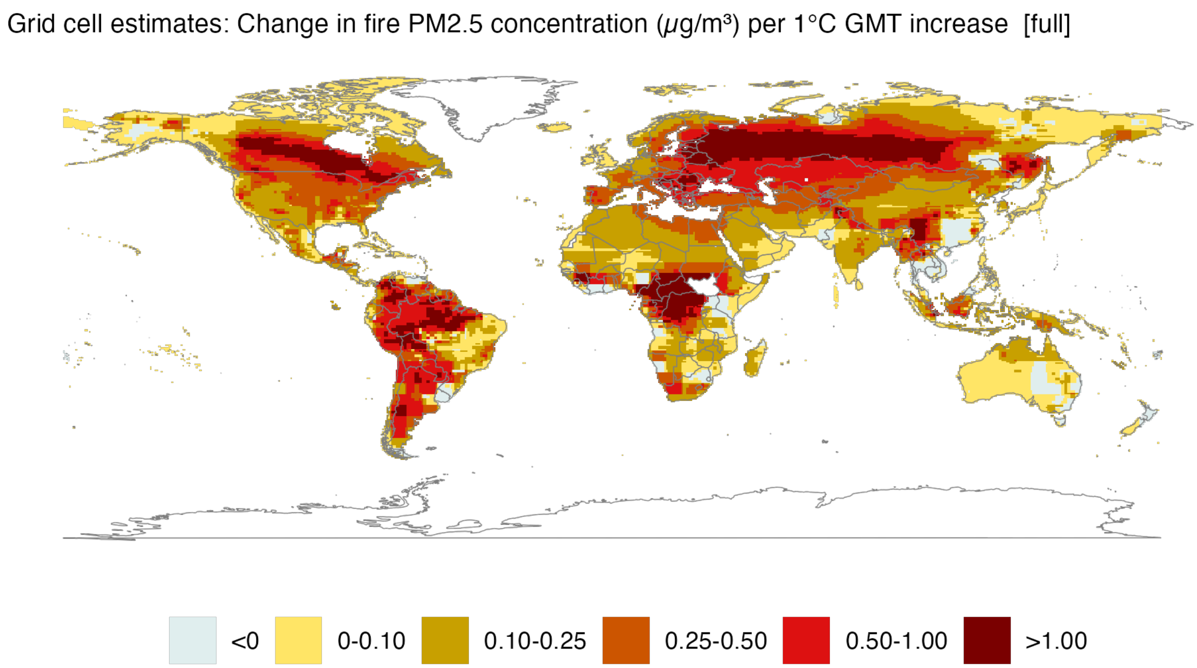


Figure 1: Grid-cell estimates of $\beta_i$, the change in wildfire $PM_{2.5}$ concentration ($\mu$g/m$^3$/yr) per 1°C increase in GMT, estimated from a grid-cell-level fixed-effects regression pooling data from Park et al. (2024), Pierce et al. (2017), and Zhao et al. (2025).

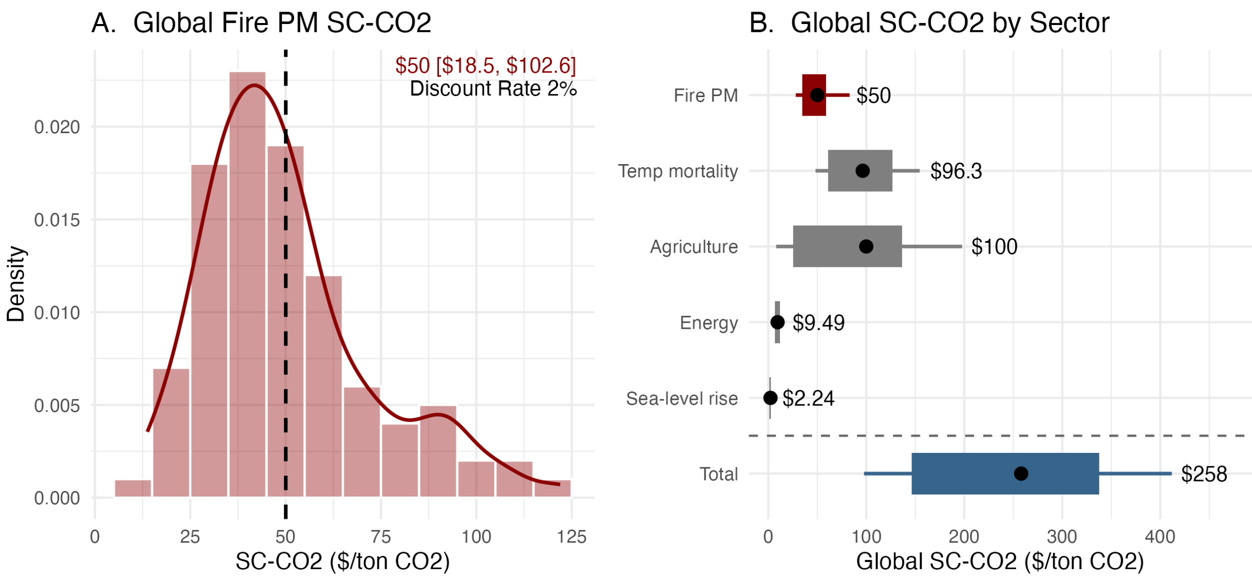


Figure 2: Global SC-$CO_2$ estimates at a 2% discount rate from 100 Monte Carlo simulations. (A) Distribution of the wildfire $PM_{2.5}$ SC-$CO_2$ component; the dashed line marks the mean ($50/ton $CO_2$; 95% conditional interval: $18.5–$102.6). (B) SC-$CO_2$ by damage sector; the dot marks the mean, the box spans the 25th–75th percentile, and whiskers span the 10th–90th percentile.

## 2 Discussion

Our key contribution is to estimate a global SC-$CO_2$ incorporating damages from wildfire smoke mortality, which no prior studies have provided. We estimate that adding this fifth damage sector increases the global SC-$CO_2$ by 24%, from $208 to $258 per ton $CO_2$ in the GIVE integrated assessment model. At $50/ton $CO_2$, wildfire smoke mortality is the third-largest damage sector in GIVE, representing a substantial source of damages missing from existing global SC-$CO_2$ estimates. Our global study complements Qiu et al. (2026)'s finding that the partial U.S. SC-$CO_2$ from wildfire smoke mortality is $11.20. Our own analysis finds a partial U.S. SC-$CO_2$ of 5.86 (Section 9). This suggests that current estimates of the SC-$CO_2$ guiding policymakers substantially understate the social cost of greenhouse gas emissions and the benefit of climate mitigation.

Our estimates are subject to several limitations. First, the concentration-response coefficient is drawn from a meta-analysis of long-term all-source $PM_{2.5}$ exposure in Europe, North America, and Asia (Pope et al., 2020), and is applied uniformly across all countries, abstracting from regional heterogeneity in population susceptibility and background pollution levels. Second, the GMT-wildfire $PM_{2.5}$ slope is identified from only 43 observations per country using wildfire $PM_{2.5}$ output from just five ESMs. This allows us to estimate a linear relationship between GMT and $PM_{2.5}$ exposure, but it does not allow us to test for curvature or estimate how the relationship changes over time. Third, our Monte Carlo intervals are conditional on the linear functional form, the set of source studies, and the assumption that model fixed effects absorb cross-model differences in levels rather than slopes. Our analysis cannot capture the full structural uncertainty in the GMT-$PM_{2.5}$ relationship, and we discuss these limitations in detail in Sections 3.4 and 8.

Finally, this paper quantifies just one cost of climate-induced wildfires: $PM_{2.5}$ mortality. Wildfire smoke contributes to non-fatal cardiovascular and respiratory disease (Johnston et al. 2021), and people are willing to pay to avoid these health impacts (Jones 2018). Wildfire smoke also appears to sharply reduce labor market earnings (Borgschulte et al. 2024). Beyond $PM_{2.5}$ emissions, wildfires generate substantial greenhouse gas emissions (Byrne et al. 2024), so wildfires may create climate tipping points (Davis et al. 2019). Moreover, wildfires damage homes and reduce property values Bayham et al., 2022. It is difficult to quantify all the costs of wildfires, but in the United States, research suggests that the total cost of wildfires is at least 1.7- 2.5 times the cost of wildfire smoke mortality (Thomas et al. 2017).[4] As a result, our estimates provide a lower-bound estimate of wildfires' impact on the social cost of carbon.

# 3 Methods

## 3.1 GMT change and wildfire smoke exposure

Together, Pierce et al. (2017), Park et al. (2024), and Zhao et al. (2025) provide globally gridded data for 43 simulations of long-term wildfire $PM_{2.5}$ exposure. Park et al. (2024) models climate-induced wildfire smoke for six decades from the 1960s to 2010s across three ESMs: Classic, Jules, and SSib4. Pierce et al. (2017) models wildfire smoke in the 2040s and 2090s along two climate pathways using CESM. Zhao et al. (2025) models wildfire smoke in the 2090s along two climate pathways using CESM2, CESM2-WACCM, EC-Earth3-CC, and EC-Earth3Veg.. We combine these data to estimate the country-level relationship between wildfire $PM_{2.5}$ exposure

[4] Thomas et al. review the literature on wildfire costs, which they summarize in Table 4.3. According to their low estimates, excess deaths from $PM_{2.5}$ mortality account for 40% of total costs; according to their high estimates, they account for 58% of total costs. Thomas et al. (2017) do not consider the effect of smoke on productivity or the cost of greenhouse gas emissions.

and global mean temperature (GMT) change.[5] We estimate this regression for each country $c$:

$$\bar{PM}_{ctsm} = \beta_c T_{ts} + \alpha_{cm} + \varepsilon_{ctsm}, \tag{1}$$

where $\bar{PM}_{ctsm}$ is population-weighted[6] per-capita wildfire $PM_{2.5}$ exposure ($\mu$g/m$^3$/yr) for country $c$, at time period $t$, in climate scenario $s$, and for wildfire model $m$; $T_{ts}$ is the GMT change relative to the 1850-1900 pre-industrial baseline (°C); $\alpha_{cm}$ is a country-model intercept to control for differences across ESMs;[7] and $\varepsilon_{ctsm}$ is a residual. The coefficient of interest, $\beta_c$, captures the change in population-weighted per-capita wildfire $PM_{2.5}$ exposure associated with a 1°C increase in GMT for country $c$. We assume that this relationship is linear and constant over time[8].

We estimate equation (1) at two spatial scales. At the country level, the dependent variable is population-weighted per-capita wildfire $PM_{2.5}$ exposure $\bar{PM}_{ctsm}$, yielding the country-specific coefficient $\beta_c$ that enters the GIVE damage function. At the grid-cell level, the dependent variable is grid-cell wildfire $PM_{2.5}$ exposure $PM_{itsm}$, yielding a grid-cell-specific coefficient $\beta_i$, the estimated change in wildfire $PM_{2.5}$ concentration per 1°C increase in GMT at grid cell $i$. The latter estimates are used solely to characterize the spatial pattern of the GMT-fire $PM_{2.5}$ relation (Figure 1).

The dependent variable is constructed by combining three sources of gridded wildfire $PM_{2.5}$ data (Section 4.1), each paired with its corresponding GMT value. Park et al. (2024) provide historical decadal mean wildfire $PM_{2.5}$ for the 1960s through 2010s under three wildfire

[5]We interpret GMT as a policy-relevant index of global climate change rather than as the direct physical driver of wildfire $PM_{2.5}$. wildfire activity and smoke exposure respond mechanistically to regional temperature, vapor pressure deficit, precipitation, fuel availability, vegetation, ignition, suppression, land use, and atmospheric transport (Park et al., 2024; Pierce et al., 2017; Wimberly et al., 2024; Zhao et al., 2025). Therefore, $\beta_c$ should not be interpreted as the causal effect of GMT on wildfire $PM_{2.5}$, but as a reduced-form mapping from GMT to the wildfire-$PM_{2.5}$ exposure implied by the available model ensemble. This parameterization is appropriate given the limited availability of globally harmonized wildfire-$PM_{2.5}$ projections and allows us to construct a country-level damage-function parameter for GIVE, where GMT is the state variable linking emissions to climate damages.

[6]See Section 4.3 for population-weighting detail.

[7]We assume each model has general assumptions that lead to level differences in GMT-wildfire smoke relationships. Our approach controls for these differences and uses within-model variation to evaluate the relationship betweeen GMT and wildfire smoke.

[8]Qiu et al. (2026) estimate a quadratic damage function relating GMST to wildfire-smoke mortality rates in the United States. A more flexible functional form is appropriate in their setting because their analysis is U.S.-specific and uses a much richer simulation design: they combine an empirically estimated North American climate-fire-smoke model with projections from 28 CMIP6 climate models, yielding 784 climate simulations spanning approximately 0.95–6.93°C of warming. By contrast, our objective is to estimate country-specific GMT-fire $PM_{2.5}$ exposure relationships globally using harmonized output from a small number of heterogeneous wildfire $PM_{2.5}$ modeling studies. For most countries, this provides relatively few distinct warming levels and limited within-model variation with which to identify curvature. We therefore impose a linear specification as a parsimonious approximation to the GMT-fire $PM_{2.5}$ relationship. This avoids fitting curvature that is weakly identified by our data and avoids imposing stronger assumptions about high-warming extrapolation. In this context, the linear specification is also conservative relative to convex specifications.

models (CLASSIC, JULES, SSiB4) and two scenarios (factual and counterfactual), yielding 36 observations per country (3 models × 2 scenarios × 6 decades). Pierce et al. (2017) provide projections for a baseline period (2001–2010) and two future periods (2041-2050, 2091-2100) under RCP4.5 and RCP8.5 using the CESM wildfire model, contributing 5 observations. Zhao et al. (2025) provide two additional projections for 2095-2099 under SSP2-4.5 and SSP5-8.5, for a total of 43 observations per grid cell and country. These gridded wildfire $PM_{2.5}$ exposure data are aggregated to population-weighted per-capita country-level exposures.

The five wildfire models differ systematically in their absolute estimates of $PM_{2.5}$ exposure due to differences in model structure rather than the GMT-fire relationship of interest in this paper. Without accounting for these differences, we would risk attributing to GMT variation in exposure levels that instead reflects differences across wildfire models. We address this by including country-model fixed effects $\alpha_{cm}$, which absorb mean model-specific exposures, so that $\beta_c$ is identified from within-model covariation between GMT and wildfire smoke across periods and scenarios.[9] This specification estimates a pooled within-model slope, $\beta_c$, across the five ESMs. If wildfire models differ not only in their mean exposure levels but also in their GMT-fire $PM_{2.5}$ slopes, then $\beta_c$ should be interpreted as an average slope. Conventional regression standard errors capture residual variation around this pooled slope, but they do not separately identify between-model slope heterogeneity as a distinct source of structural uncertainty (Section 3.4).

The resulting $\beta_c$ estimate is used in the GIVE integrated assessment model as the country-level damage function parameter linking GMT change to wildfire smoke exposure and its associated mortality. Across 174 countries, $\beta_c$ is positive for 93% of countries (median = 0.20 $\mu$g/m$^3$/yr per °C), indicating that warming broadly increases population exposure to wildfire $PM_{2.5}$. Estimates are statistically significant at the 5% level for 65% of countries. The largest effects are concentrated in northern North and South America, northern Asia, central Africa, and parts of Southeast Asia, while fewer countries—predominantly in Africa as well as Australia—have negative $\beta_c$ (Figure 3). For Australia, negative relationships at the grid-cell level in the most populous regions of the country (Figure 1) translate into declining population-weighted exposure with increasing GMT.

## 3.2 Wildfire smoke exposure and mortality

Defining a wildfire smoke damage function in GIVE requires a parameter relating $PM_{2.5}$ exposure and mortality. Following Qiu et al. (2025) and Qiu et al. (2026) [10], we source this parameter

[9] Figure 5 in the Appendix illustrates this for select countries.

[10] These studies use the all-source $PM_{2.5}$ Pope et al. (2020) result for exposure levels outside historical observations in the U.S., those $> 6$ $\mu$g/m$^3$, while using their own wildfire $PM_{2.5}$-mortality response estimate for exposure $< 6$ $\mu$g/m$^3$. We instead apply the Pope et al. (2020) result for all exposure levels. Since there is evidence of

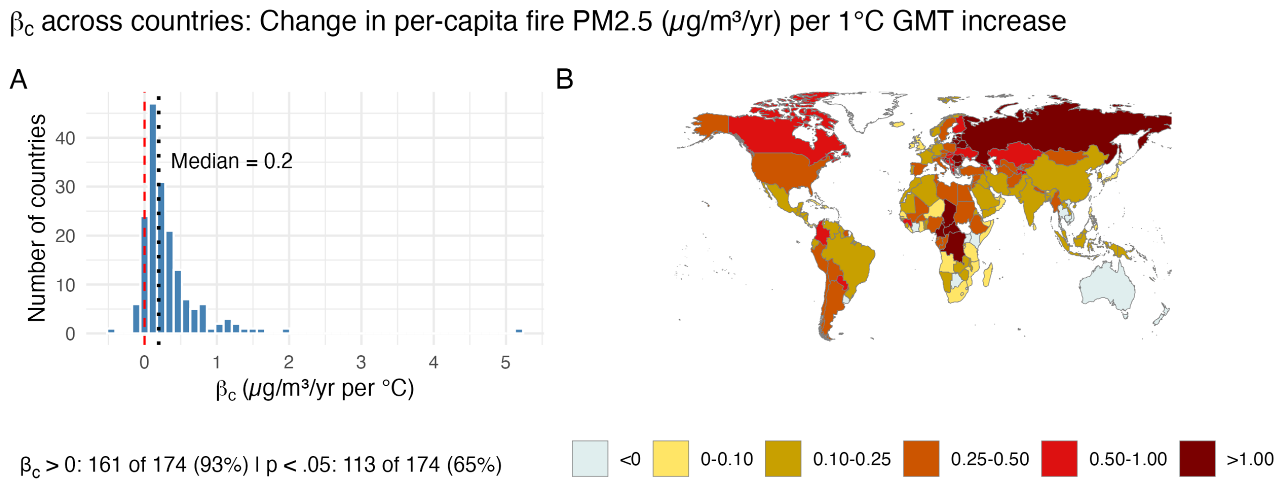


Figure 3: Country-level estimates of $\beta_c$, the change in population-weighted per-capita wildfire $PM_{2.5}$ exposure ($\mu g/m^3/yr$) per 1°C increase in GMT. (A) Distribution of $\beta_c$ across 174 countries; the dashed red line marks zero and the dotted black line marks the median (0.20 $\mu g/m^3/yr$ per °C). (B) Heat map of $\beta_c$ by country.

from the Pope et al. (2020) meta-analysis of 33 selected studies in Europe, North America, and Asia of long-term exposure to $PM_{2.5}$. Using a random effects model, they estimate a hazard ratio (HR) of 1.08 (95% CI =1.06-1.11) for all-cause mortality per 10 $\mu g/m^3$. This estimate is a bit lower than the Orellano et al. (2024) meta-analysis, which reports a central estimate of 1.095 (95% CI: 1.064–1.127) per 10 $\mu g/m^3$. We use the Pope et al. (2020) estimate to calculate the percent increase in all-cause mortality rates per $\mu g/m^3$ as $\gamma = \frac{1.08-1}{10} = 0.008$. This 0.8% increase per $\mu g/m^3$ is applied to all countries globally.

## 3.3 GIVE wildfire smoke damage function

Our GIVE damage function takes a linear form analogous to the existing temperature-related mortality module in Rennert et al. (2022). We translate warming-induced wildfire $PM_{2.5}$ exposure into monetized mortality damages similarly. For country $c$ at time $t$, monetized damages are

$$C_{ct} = V_{ct} \cdot M_{ct} \cdot \gamma \cdot \beta_c \cdot T_t, \tag{2}$$

where $V_{ct}$ is the value of a statistical life (VSL) in USD, $M_{ct}$ is the baseline all-cause mortality count, $\gamma = 0.008$ is the concentration-response coefficient (the proportional increase in mortality risk per 1 $\mu g/m^3$ increase in long-term $PM_{2.5}$ exposure), $\beta_c$ is the country-specific regression coefficient estimated in equation (1) (the change in population-weighted per-capita wildfire $PM_{2.5}$ exposure per 1°C increase in GMT), and $T_t$ is the GMT change relative to the pre-

greater toxicity from wildfire $PM_{2.5}$ (Qiu et al., 2025), we believe our estimated impact to be conservative.

industrial baseline (°C).

The product $M_{ct} \cdot \gamma \cdot \beta_c \cdot T_t$ gives the number of excess deaths attributable to warming-induced wildfire $PM_{2.5}$ in country $c$ at time $t$. Multiplying by $V_{ct}$ converts these excess deaths to monetary damages. The VSL is sourced from (Rennert et al., 2022): a baseline U.S. VSL for 2020 is adjusted for each country and year using GDP per capita with an income elasticity of one. The baseline all-cause mortality count, $M_{ct}$, is likewise sourced from (Rennert et al., 2022), which projects these values for country and year from 2025 to 2300.

## 3.4 Parameter sampling and sensitivity analysis

Following Rennert et al. (2022), GIVE propagates uncertainty through Monte Carlo simulation. In the baseline GIVE model, each simulation samples uncertain socioeconomic and emissions trajectories (RFF-SPs), climate model parameters (FaIR), sea-level model parameters (BRICK), and damage-function coefficients. We retain this existing GIVE uncertainty structure and add parameter sampling for the two quantities specific to the wildfire $PM_{2.5}$ damage sector: the concentration-response coefficient $\gamma$ and the country-level GMT-exposure coefficient $\beta_c$.

For $\gamma$, we draw from a normal distribution with mean $\bar{\gamma} = 0.008$ and standard error $SE(\gamma) = 0.00128$, derived from the 95% confidence interval of the Pope et al. (2020) hazard ratio (HR = 1.08; 95% CI: 1.06–1.11 per 10 $\mu$g/m$^3$), converted to a per-$\mu$g/m$^3$ mortality response as $\gamma = (HR - 1)/10$.[11] For $\beta_c$, we draw independently for each country from a normal distribution with mean $\hat{\beta}_c$ and standard error $SE(\hat{\beta}_c)$ from the country-level fixed-effects regression in equation 1. These draws generate a distribution of wildfire smoke mortality damages and associated SC-$CO_2$ estimates conditional on our preferred damage-function specification.

These simulations should be interpreted as a parameter-sensitivity analysis rather than a full characterization of uncertainty. The resulting intervals reflect uncertainty in the concentration-response coefficient and sampling uncertainty in the estimated country-level GMT-$PM_{2.5}$ coefficients, conditional on the source studies used, the harmonization procedure, the linear specification, the country-model fixed-effects pooling structure, and the global $PM_{2.5}$ mortality coefficient. They do not fully propagate uncertainty from alternative wildfire models, between-model heterogeneity in GMT-$PM_{2.5}$ slopes, alternative functional forms, regridding choices, regional heterogeneity in smoke toxicity or population vulnerability, future adaptation, or changes in within-country population exposure. We therefore report these intervals as conditional Monte Carlo sensitivity intervals, not as full structural uncertainty bounds.

To evaluate the sensitivity of the results to model structure, we also compare our preferred estimate to estimates excluding individual wildfire models. This comparison is reported separately from the Monte Carlo parameter sampling because the small number of source models

[11] Specifically, $SE(\gamma) = (\gamma_{upper} - \gamma_{lower})/(2 \times 1.96) = (0.011 - 0.006)/3.92 \approx 0.00128$.

does not support a formal decomposition of between-model structural uncertainty. Together, the Monte Carlo simulations and model-specific comparisons indicate how the wildfire smoke SC-$CO_2$ varies under our maintained assumptions, while also highlighting the sources of uncertainty that remain outside the reported intervals.

# 4 Appendix: Data

## 4.1 Wildfire smoke data and regridding

Pierce et al. (2017), Park et al. (2024), and Zhao et al. (2025) each generate global gridded wildfire $PM_{2.5}$ datasets and project how climate change alters wildfire activity and associated concentrations, but differ in their temporal coverage, methodology, and treatment of non-climate drivers. All studies simulate gridded emissions estimates from wildfire, which are fed into chemical transport models to produce wildfire $PM_{2.5}$ contributions to total $PM_{2.5}$. The wildfire $PM_{2.5}$ contributions are isolated as the difference between simulations with and without wildfire emissions conducted under the same scenario. GMT change is a key driver of wildfire $PM_{2.5}$ outcomes in all studies. However, another key factor that varies in all studies is land cover and land use (LCLU), a variable capturing deforestation. Although Pierce et al. (2017, p. 11) state it is "obvious that future wildfire pollution will be primarily driven by climate," LCLU changes also impact outcomes.

Pierce et al. (2017) use Community Earth System Model (CESM)—a coupled chemistry-climate-land model—to simulate wildfire activity and $PM_{2.5}$ concentrations for a baseline period (2001–2010) and future periods (2041–2050 and 2091–2100) under RCP4.5 and RCP8.5. Their simulation design isolates the effect of climate change on wildfire activity by including both scenarios in which population density varies with SSP trajectories and scenarios in which it is held constant at year-2000 levels. In the chemical transport step, anthropogenic and biogenic emissions are likewise held constant within each snapshot period. We use only the population-constant simulations, as our regression targets the GMT-fire $PM_{2.5}$ relationship excluding the impact of population change (Section 3.1).

Park et al. (2024) simulate historical wildfire $PM_{2.5}$ from the 1960s through the 2010s using three ESMs (CLASSIC, SSiB4, and JULES) driven by historically observed climate forcings. In the chemical transport model, anthropogenic emissions are set to observed historical values for each decade from Community Emissions Data System Inventory (CEDS) and meteorological conditions are fixed at a single reference year, to attribute to climate change the variation in wildfire $PM_{2.5}$ across decades.[12] Their climate change attribution involves a factual scenario (simulating observed outcomes) and a counterfactual scenario (simulating outcomes in a world without warming). We use the factual and counterfactual simulations, which span six decadal means across three wildfire models, yielding 36 observations per country.

Zhao et al. (2025) develop a machine learning model of burned area trained on Global Fire Emissions Database version 4.1s (GFEDv4.1s) observed burned area, ERA5 climate reanalysis, and Earth System Model (ESM) estimates. The Coupled Model Intercomparison Project

[12]Specifically, Park et al. (2024) fix GEOS-Chem meteorological conditions at 2016 values for all simulation periods (Table 1 of Park et al. (2024)).

Phase 6 (CMIP6)-projected climate variables are used to predict future global wildfire gridded emissions estimates for 2095–2099 under SSP2-4.5 and SSP5-8.5. These emissions enter into the GEOS-Chem chemical transport model to produce wildfire $PM_{2.5}$ concentrations. Land-use inputs follow scenario-specific Land Use Harmonization 2 (LUH2) trajectories, so land use varies with the warming scenario. Anthropogenic emissions likewise vary with the SSP scenario.

We regrid $PM_{2.5}$ data from these multiple sources and native resolutions onto a unified $0.5° \times 0.5°$ global grid ($720 \times 360$), and subsequently merge the gridded fields with population data using longitude–latitude matching. The only difference across the three scripts lies in the resampling method: the Park et al. (2024) dataset is interpolated using bilinear interpolation, the Pierce et al. (2017) dataset is remapped using nearest-neighbor matching, and the Zhao et al. (2025) dataset is aggregated using area-weighted averaging. All other processing steps are identical, including the definition of the target grid, coordinate harmonization, and the final left join with population data based on exact longitude-latitude pairs. The resulting grid is defined at the center of each 0.5° cell, with longitude spanning $-179.75°$ to $179.75°$ and latitude spanning $-89.75°$ to $89.75°$, at 0.5° intervals.

## 4.2 Global mean temperature change

Global mean temperature change values relative to pre-industrial times are associated with each $PM_{2.5}$ simulation. For the Park et al. (2024) simulations, we assign historically observed GMT changes (OWID, 2026). For Zhao et al. (2025), we assign GMT changes projected under SSP2-4.5 and SSP5-8.5 (Anthoff, 2026). For Pierce et al. (2017), we use their gridded GMT change data relative to 2005, to which we apply a pre-industrial offset (OWID, 2026).

## 4.3 Population weighting

For grid cell $i$ in country $c$, let $PM_{itsm}$ denote wildfire $PM_{2.5}$ exposure ($\mu$g/m$^3$/yr) under time period $t$, scenario $s$, and wildfire model $m$. Each observation is already a period average: a decade mean for Park et al. (2024) and Pierce et al. (2017), and a five-year mean for Zhao et al. (2025).

Baseline population $\bar{pop}_{ic}$ is the average annual population in cell $i$ of country $c$ over 2001-2010, sourced from Pierce et al. (2017). Population is held fixed at this baseline across all periods and scenarios to isolate changes in wildfire $PM_{2.5}$ concentrations. Country-level baseline population is $\bar{pop}_c = \sum_i \bar{pop}_{ic}$.

Population-weighted per-capita wildfire $PM_{2.5}$ exposure for country $c$, period $t$, scenario $s$, and wildfire model $m$—the dependent variable in equation (1)—is:

$$\bar{PM}_{ctsm} = \frac{\sum_i PM_{itsm} \cdot p\bar{o}p_{ic}}{p\bar{o}p_c}.$$

# 5 Appendix: Regression descriptive statistics

Figure 4 summarizes the distribution of the two regression inputs across the 7,482 country-period-scenario-model observations used in equation (1). Panel A shows the cross-country distribution of population-weighted per-capita wildfire $PM_{2.5}$ exposure ($\mu$g/m$^3$/yr) by wildfire model. Pooled observations across all models, the distribution is heavily right-skewed: the median is 0.54 while the mean is 1.79, and the 90th percentile (4.75) is roughly seventy times the 10th percentile (0.08). Mean exposure differs substantially across wildfire models—ranging from 1.02 (JULES) to 2.56 (CLASSIC)—motivating the inclusion of wildfire-model fixed effects to absorb these level differences before estimating the GMT change-exposure slope. CESM exposure levels are generally higher than the Park and Zhao models across the distribution: the CESM 25th percentile (0.77) exceeds the median of all three Park models (0.35–0.51) and the Zhao median (0.64).

Panel B shows the discrete GMT change values (°C relative to the pre-industrial) assigned to each data source. The six Park et al. (2024) historical decade means span 0.25–1.09°C for factual observations, with 0°C warming for counterfactual observations, the low end of the temperature range. The five Pierce et al. (2017) projections extend from 1.04°C (baseline period) to 4.47°C (RCP8.5, 2091–2100), and the two Zhao et al. (2025) projections contribute values of 2.92°C (SSP2-4.5) and 4.96°C (SSP5-8.5). Together the three sources span approximately 0–5.0°C, enabling identification of $\beta_c$ across a wide range of warming levels.

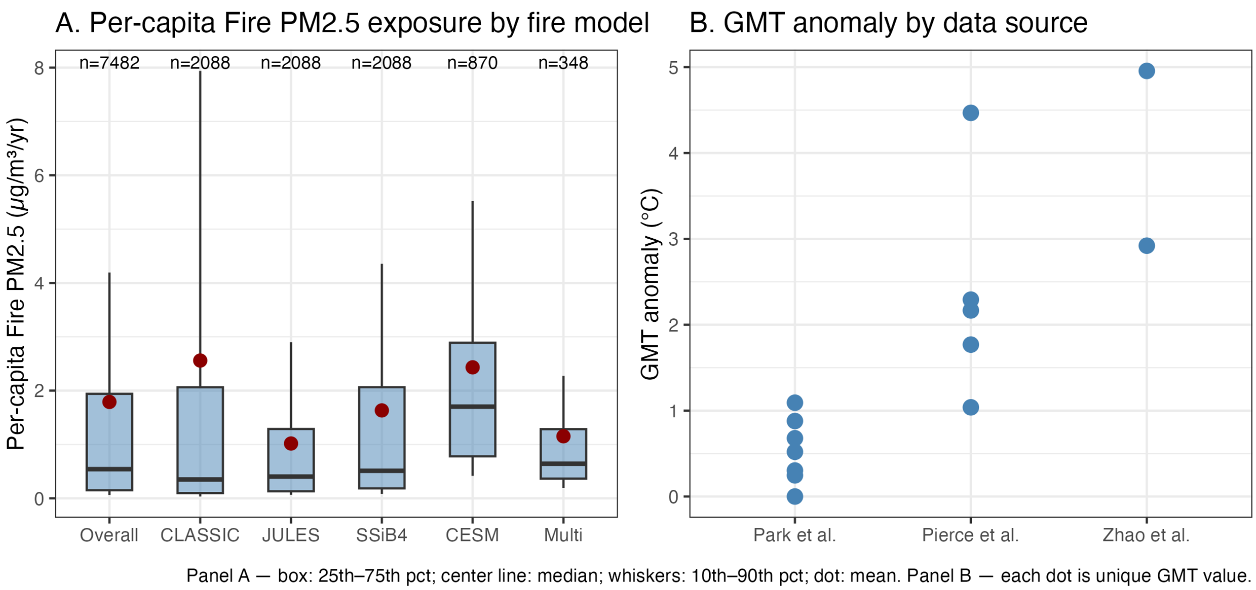


Figure 4: Distribution of regression inputs. (A) Cross-country distribution of population-weighted per-capita wildfire $PM_{2.5}$ exposure ($\mu g/m^3/yr$) for the full sample and by wildfire model; box spans the 25th–75th percentile, center line is the median, whiskers span the 10th–90th percentile, and the red dot marks the mean. (B) Discrete GMT change values (°C relative to the 1850–1900 pre-industrial) used as regressors, by data source; each dot is one unique $T_{ts}$ value.

# 6 Appendix: Regression results

Figure 5 plots per-capita wildfire $PM_{2.5}$ exposure by GMT change for the five most populous countries as well as the global average. The plots show the common estimated slope, $\beta_c$, as well as the unique fixed-effect intercepts for each model. Note that $\beta_c$ is estimated for 174 countries. We also estimate $\beta$ globally, which is applied to select small countries captured in the GIVE model for which our country list did not have a match. The CESM model tends to have higher exposure than other models and has the greatest variation in GMT change. In leave-one-out sensitivity tests we estimate $\beta_c$ with data from one model excluded (Section 10).

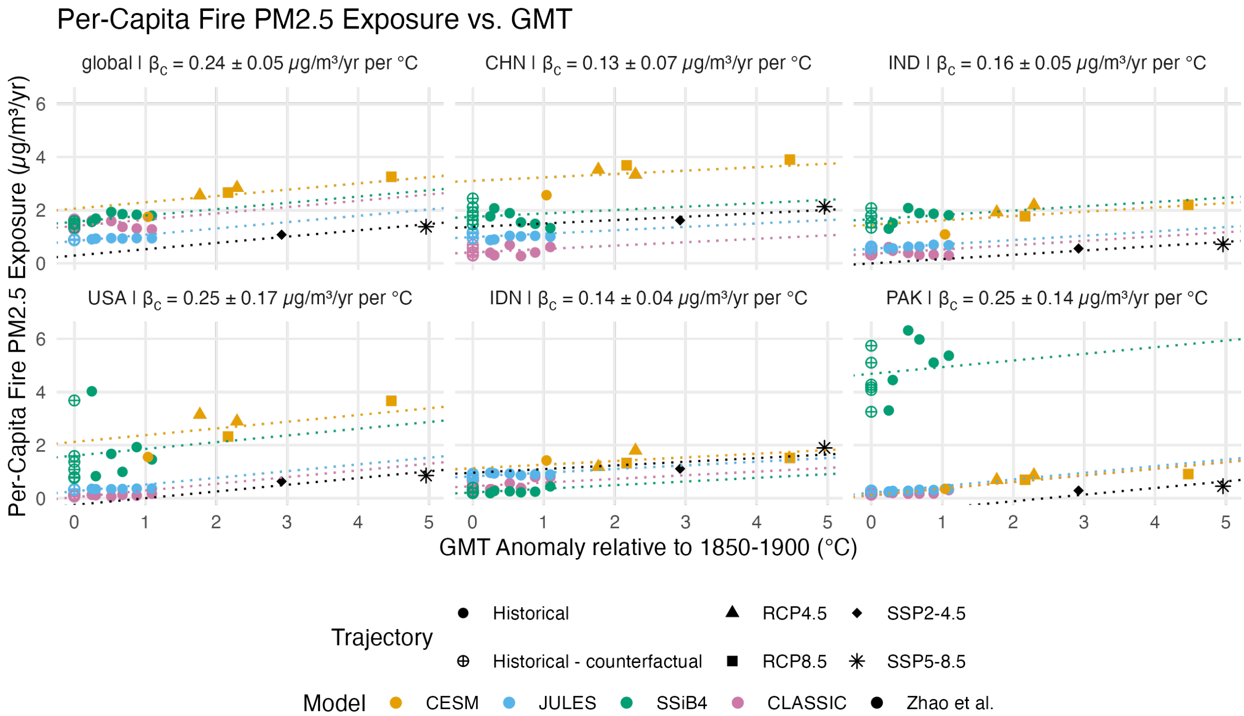


Figure 5: Per-capita wildfire PM2.5 exposure ($\mu$g/m$^3$/yr) by GMT change relative to the pre-industrial baseline (°C) for the five largest population countries as well as the global average. Each plot shows 43 observations from Park et al. (2024), Pierce et al. (2017), and Zhao et al. (2025). Point color indicates wildfire model; point shape indicates warming trajectory. The five dotted lines show the fitted line for each wildfire model, using the common slope $\beta_c$ and the unique fixed-effect intercept.

# 7 Appendix: FUND region decomposition

Table 1 decomposes the mean SC-$CO_2$ by FUND region and damage sector. The largest wildfire $PM_{2.5}$ components are concentrated in Sub-Saharan Africa (\$11.42/t$CO_2$), the Former Soviet Union (\$8.73/t$CO_2$), the United States (\$5.85), Middle East (\$4.11), and Western Europe (\$2.56)—regions combining large positive $\beta_c$ estimates, substantial exposed populations, and higher VSLs. Australia and New Zealand exhibit a positive wildfire $PM_{2.5}$ contribution (\$0.12), despite negative $\beta_c$ estimates for those countries (Figure 3), due to large standard errors and random sampling. The largest total SC-$CO_2$ values are in South Asia (\$44.93), Middle East (\$40.23), and Sub-Saharan Africa (\$39.68), driven primarily by temperature mortality and agriculture damages.

The regional decomposition highlights a notable feature of studies applying a VSL that varies by country according to GDP per capita: Given the same physical damage (fire $PM_{2.5}$ and temperature mortality), poorer countries will have a lower monetized damage than rich countries. This means that damages in South Asia or Sub-Saharan Africa are substantially discounted relative to those in the United States or Western Europe for mortality impacts that are comparable or even larger. The equity implications of VSL-based damage valuation are an important issue beyond the scope of this paper.

Table 1: SC-CO2 by FUND Region ($/tCO2)

| Fund Region | Total | Fire PM | Temp mortality | Agriculture | Energy | Sea-level rise |
|---|---|---|---|---|---|---|
| Sub-Saharan Africa | 39.68 | 11.42 | 11.58 | 12.90 | 3.67 | 0.10 |
| Former Soviet Union | 21.03 | 8.73 | 4.46 | 7.74 | 0.04 | 0.06 |
| United States | 21.67 | 5.85 | 13.23 | 1.59 | 0.62 | 0.38 |
| Middle East | 40.23 | 4.11 | 17.61 | 17.09 | 1.34 | 0.07 |
| Western Europe | 7.57 | 4.03 | 2.56 | 0.72 | 0.10 | 0.16 |
| South Asia | 44.93 | 3.66 | 9.97 | 30.14 | 0.99 | 0.16 |
| Latin America | 8.75 | 2.75 | 2.06 | 3.36 | 0.37 | 0.22 |
| China | 29.74 | 2.44 | 13.14 | 13.40 | 0.65 | 0.11 |
| North Africa | 16.86 | 1.81 | 7.26 | 7.16 | 0.58 | 0.05 |
| Eastern Europe | 3.32 | 1.66 | 1.25 | 0.38 | 0.01 | 0.02 |
| Canada | 1.92 | 1.32 | 1.31 | -0.70 | -0.00 | 0.00 |
| Southeast Asia | 13.15 | 1.12 | 6.39 | 4.32 | 0.90 | 0.43 |
| Central America | 3.95 | 0.60 | 0.86 | 2.29 | 0.13 | 0.06 |
| Japan, Korea | 4.25 | 0.35 | 3.26 | 0.52 | 0.04 | 0.09 |
| Australia, New Zealand | 0.57 | 0.12 | 1.16 | -1.00 | 0.02 | 0.27 |
| Small Island States | 0.46 | 0.06 | 0.23 | 0.07 | 0.03 | 0.06 |
| Total | 258.08 | 50.03 | 96.33 | 99.98 | 9.49 | 2.24 |

# 8 Appendix: Qiu et al. (2026) method

## 8.1 Summary of Qiu et al. (2025) and Qiu et al. (2026)

Using a data-driven approach focused on the USA, Qiu et al. (2025) model the full causal chain linking GMT to wildfire smoke emissions, wildfire smoke emissions to smoke concentrations, and smoke concentrations to mortality. They (i) "construct an ensemble of statistical and machine learning models that predict wildfire emissions as a function of climate and land use variables over North America"; (ii) "use surface wildfire smoke $PM_{2.5}$ estimates derived from satellite data and machine learning models to establish an empirical relationship between wildfire emissions and smoke $PM_{2.5}$ concentrations across the contiguous USA at a 10-km resolution"; and (iii) use a Poisson distributed-lag model to estimate the effects of annual smoke $PM_{2.5}$ on county-level all-cause mortality rates from 2006-2019 (Qiu et al., 2025, p. 936). Qiu et al. (2026) use this framework to estimate a quadratic damage function relating GMT change to smoke-related mortality in the USA, which is incorporated into the GIVE model to compute a partial SC-$CO_2$. With "projections from 28 global climate models (GCMs) from the sixth phase of the Coupled

Model Intercomparison Project (CMIP6)," Qiu et al. (2026, p. 2) project forward their causal model to estimate future mortality. Each of the 28 GCMs have climate variable outputs for four SSPs (shared socioeconomic pathway) and seven 20-year time periods. Additionally, they develop an empirical wildfire-fuel feedback that reduces projected wildfire emissions at higher warming levels.

## 8.2 Strengths and limitations of each approach

**Our approach.** The primary strength of our approach is global coverage. Whereas Qiu et al. (2025) require 10-km resolution smoke $PM_{2.5}$ predictions for the USA, soil moisture from NLDAS-2 (North American Land Data Assimilation System)—reportedly available only in the contiguous USA—and county-level mortality data covering the contiguous United States, the datasets we use (Park et al., 2024; Pierce et al., 2017; Zhao et al., 2025) generate globally gridded wildfire $PM_{2.5}$ concentrations that can be translated into country-level damage function parameters for all GIVE regions.

As noted above, a further advantage of our approach is that the Park et al. (2024) and Pierce et al. (2017) datasets capture dynamic feedbacks between GMT change and wildfire activity. Specifically, these ESMs predict how climate change-driven wildfires will impact vegetation growth and fuel loads, which in turn influences future wildfire activity. Qiu et al. (2026) acknowledge this as an important factor missing from their approach.

The primary limitation of our approach is limited data. Whereas Qiu et al. (2026) estimate their quadratic damage function with 784 climate simulations (28 GCMs $\times$ 4 SSPs $\times$ 20 time periods), our country-level regression draws on 43 observations per country, which is sufficient to identify a linear, constant slope, but no curvature or time-temperature interactions. Additionally, our uncertainty intervals are conditional: they reflect sampling uncertainty in $\beta_c$ and $\gamma$ but do not fully capture disagreement across wildfire models in the GMT-$PM_{2.5}$ slope, or uncertainty arising from alternative functional forms (Section 3.4).

**Qiu et al.** While we rely on output from ESMs relating GMT to $PM_{2.5}$ concentrations, Qiu et al. (2025, p. 941) model "the full causal chain linking climate to health through wildfires (climate change-wildfire–air pollution–mortality), and [they propagate] uncertainty from each step to understand uncertainty in likely health impacts." As they note, "one advantage of our data-driven climate-fire model over process-based model projections is its ability to isolate the factors of climate variables from other factors that influence wildfires" (Qiu et al., 2025, p. 941). In contrast, in the Pierce et al. (2017) model, land-use change assumptions (deforestation) influence future wildfire $PM_{2.5}$ concetrations—a factor for which our regression does not control—but Pierce et al. also suggest that climate variables are the primary driver of concentrations (Ford et al., 2018). Additionally, in the Qiu et al. (2025, pp. 935–936) approach, the "magnitude of smoke PM2.5 is directly constrained by the empirical estimates of smoke PM2.5 concentrations," avoiding the reliance on chemical transport models that can yield estimates differing "by an order of magnitude when compared with surface observations." Their approach also supports identification of a quadratic GMT change-mortality relationship, and Qiu et al. (2026) incorporate an empirically estimated wildfire-fuel feedback that reduces the SC-$CO_2$ by 68% relative to a no-feedback specification. This feedback effect is not explicitly captured in our regression but is accounted for in our $PM_{2.5}$ simulation data (Park et al., 2024; Pierce et al., 2017), as noted above.

The primary limitation of Qiu et al. (2026) is geographic scope. Their data requirements—particularly for satellite-derived smoke $PM_{2.5}$ at 10-km resolution and NLDAS-2 soil moisture—seem to confine the approach to North America, preventing global application. As noted above, an additional limitation is not fully capturing climate-vegetation dynamics or lightning ignitions, both of which process-based models represent explicitly (Pierce et al., 2017).

## 8.3 Why Qiu et al. (2026) estimate stronger wildfire smoke effects

Several modeling aspects of Qiu et al. (2025) and Qiu et al. (2026) result in higher excess mortality estimates at higher warming levels relative to our results (Section 9).

*Climate change-$PM_{2.5}$ concentrations relation.* Qiu et al. (2026) find that 1.5°C warming increases population-weighted $PM_{2.5}$ exposure by 0.84 $\mu$g/m$^3$ in the USA, while warming of 2, 3, and 4°C increases exposure by 1.2, 2.4, and 4.5 $\mu$g/m$^3$, respectively. By contrast, our linear regression finds that population-weighted $PM_{2.5}$ exposure in the USA increases 0.25 $\mu$g/m$^3$ per 1°C warming (Section 6). Assumed constant, our estimated relation yields lower exposure at higher warming levels as well.

*Smoke-specific concentration-response function.* Qiu et al. (2025) empirically estimate a smoke-specific exposure-response function, which they find to have a larger impact on all-cause mortality than all-source $PM_{2.5}$. Their results are larger than those used in prior work, including our application of Pope et al. (2020): a 0.8% increase in all-cause mortality per $\mu$g/m$^3$ of *all-source* $PM_{2.5}$ exposure, because smoke $PM_{2.5}$ may have distinct chemical composition and toxicity. Qiu et al. (2025) state that this higher estimated exposure-response largely accounts for their higher mortality estimates relative to other studies.

*Distributed-lag mortality.* Qiu et al. (2025, p. 940) use a distributed-lag model and find that "exposure to annual mean smoke concentrations of 0.75-1 $\mu$g/m$^3$ increases the mortality rate by 1.5% in the same year, but by 2.7% ... if we also account for deaths occurring in the following year." Qiu et al. (2026) adopt this approach, accounting for exposure in the current and prior year. Our approach applies a single-period all-source response coefficient and does not capture this multi-year mortality effect.

*Quadratic damage function.* Qiu et al. (2026) estimate a quadratic GMT-mortality relationship from 784 simulations spanning 0.95-6.93°C of warming. As shown in our comparison (Table 3), the quadratic function exceeds our linear function *in levels* above approximately 3.19°C, producing substantially more projected deaths at high warming levels (e.g., 115,667 vs. 62,854 at 6°C).

# 9 Appendix: USA Results

Qiu et al. (2026) estimate a partial SC-$CO_2$ for the U.S. accounting for wildfire $PM_{2.5}$ mortality. This section compares our U.S. results against theirs. First, Figure 6 shows $\beta_i$ estimates across the contiguous United States. Values are positive throughout nearly the entire country, with the strongest responses concentrated in the northern interior—the upper Midwest and Great Lakes region—consistent with the high $\beta_c$ estimated for the United States and Canada at the country level. The weakest responses are in the Southeast and Gulf Coast. The West shows moderate-to-high values.

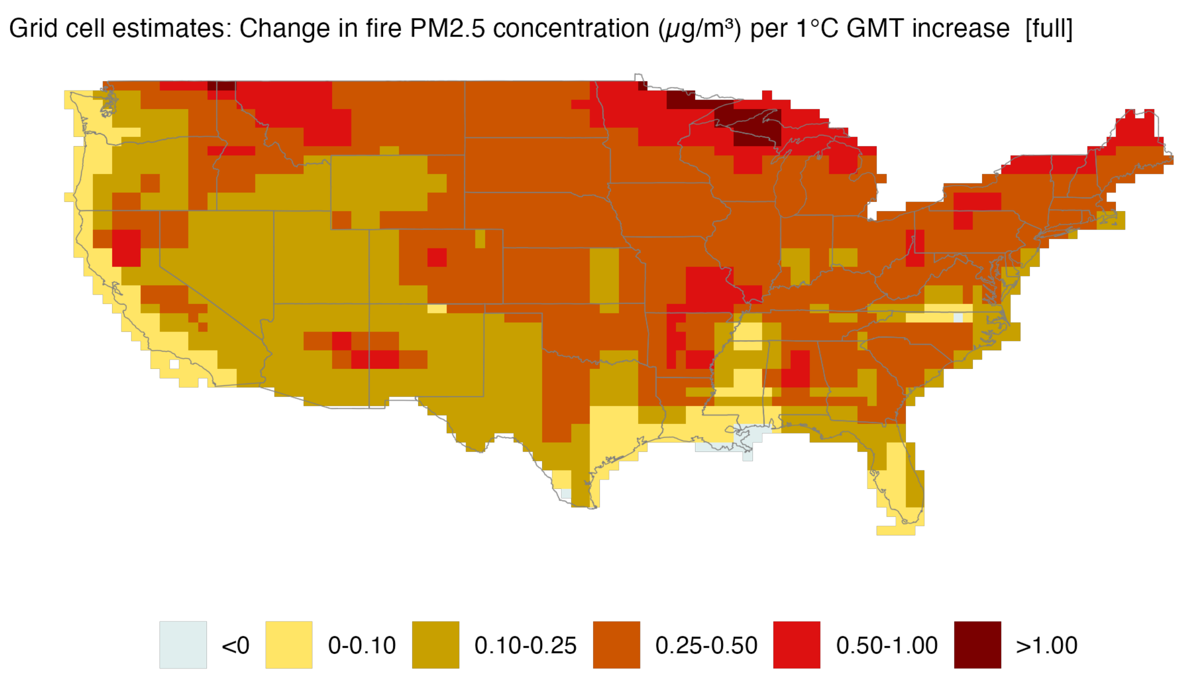


Figure 6: Grid-cell estimates of $\beta_i$, the change in wildfire $PM_{2.5}$ concentration ($\mu g/m^3/yr$) per 1°C increase in GMT, estimated from a grid-cell-level fixed-effects regression pooling data from Park et al. (2024), Pierce et al. (2017), and Zhao et al. (2025).

The U.S. wildfire $PM_{2.5}$ SC-$CO_2$ has a mean of \$5.90/ton $CO_2$ (95% conditional interval: \$-1.00–\$20.40) at a 2% discount rate (Figure 7, Panel A). Panel B shows that wildfire $PM_{2.5}$

mortality represents the second-largest damage sector after temperature mortality ($13.30) and increases the total U.S. SC-$CO_2$ by 36%, from $15.84 to $21.70. Our main U.S. result of $5.90 is below the $11.39 estimated by Qiu et al. (2026) (Table 2).[13] Differences between our results and those of Qiu et al. (2026) are driven by different damage functions, which are discussed below along with associated excess death estimates.

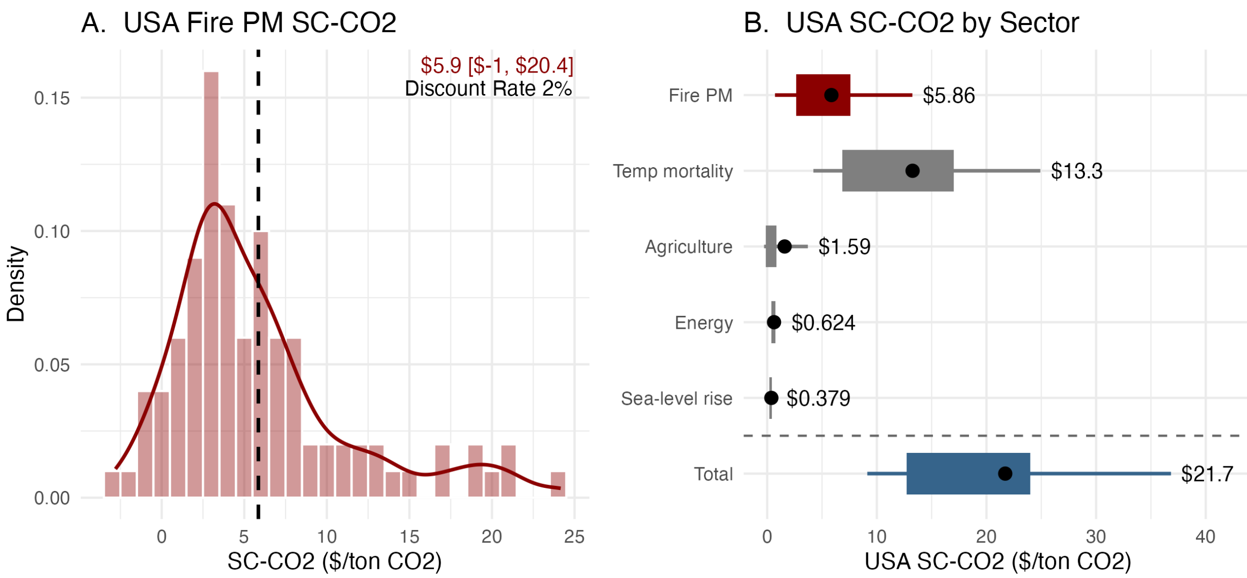


Figure 7: U.S. partial SC-$CO_2$ estimates at a 2% discount rate from 100 Monte Carlo simulations. (A) Distribution of the wildfire $PM_{2.5}$ SC-$CO_2$ component; the dashed line marks the mean ($5.90/ton $CO_2$; 95% conditional interval: $-1.00–$20.40). (B) SC-$CO_2$ by damage sector; the dot marks the mean, the box spans the 25th–75th percentile, and whiskers span the 10th–90th percentile.

Table 2: USA SC-CO2 Comparison with Qiu et al. 2026 ($/tCO2)

| Sector | SC-CO2 | SC-CO2 Qiu et al. | Difference |
|---|---|---|---|
| Temperature-related mortality | 13.26 | 13.17 | 0.09 |
| Fire PM2.5 mortality | 5.86 | 11.39 | -5.53 |
| Agriculture | 1.59 | 1.58 | 0.01 |
| Energy | 0.62 | 0.62 | 0.00 |
| Sea-level rise | 0.38 | 0.38 | 0.00 |
| Total | 21.71 | 27.14 | -5.43 |

[13] We replicate the Qiu et al. (2026) results using 100 Monte Carlo simulations rather than their published 10,000; the difference is negligible, with their published wildfire $PM_{2.5}$ SC-$CO_2$ of $11.2 closely matching our replication value of $11.39. Minor differences in the unchanged sectors (temperature mortality, agriculture, energy, and sea-level rise) reflect the fact that incorporating an additional damage sector reduces net consumption, which alters the endogenous discount rate and thus affects all sector estimates marginally.

Figure 8 compares our linear damage function to the Qiu et al. (2026) quadratic function. This is illustrative only, evaluated at a fixed U.S. population of 340 million and a baseline death rate of 8.9 per 1,000. Note also that the Qiu et al. $\beta$s are averaged across their 10,000 trial runs. Table 3 reports the actual average excess deaths produced by the GIVE model—comparing our results to Qiu et al. (2026)—where population evolves over time.

Two features of Figure 8 are important. First, the dotted vertical line marks 1.59°C: the GMT change at which the *marginal* excess deaths of the Qiu et al. quadratic first exceeds that of our linear function. Recall that the *marginal* excess death is what drives wildfire $PM_{2.5}$ SC-$CO_2$ estimates. Below 1.59°C, each additional degree of warming causes more *marginal* deaths under our function than under Qiu et al. (2026). On the one hand, warming of >1.59°C occurs over most years modeled in GIVE, which covers 2025 to 2300. This means that most years will have greater marginal damage from wildfire under Qiu et al. than under our results. On the other hand, the earlier decades (with less warming) when our marginal damage is greater corresponds to years with less discounting. In later years, each death is worth less in monetary terms, due to higher discounting.[14] This means that, although lower, our wildfire $PM_{2.5}$ SC-$CO_2$ estimates are not substantially lower than those of Qiu et al., as might be expected for the reasons outlined in Section 8.3. Second, the two curves cross in *levels* at 3.19°C, after which the Qiu et al. quadratic projects more excess deaths than our linear function.

GIVE estimates excess death results for each year, country, and iteration $n$. Table 3 reports U.S. annual excess deaths averaged across all year-iteration pairs where GMT change is ±0.10°C of each reported level. For example, the 1.5°C row averages over all year-iteration pairs with $T \in [1.4, 1.6]$. These results reflect the levels crossover shown in Figure 8. At 1.5°C our function projects 11,327 excess deaths annually compared to 1,128 for Qiu et al. (2026), with the gap remaining large at 2°C (16,604 and 7,172). The estimates cross in levels between 3–4°C, after which Qiu et al. project more excess deaths than our linear function at all higher warming levels (115,667 and 62,854 at 6°C).

Note that Qiu et al. (2026) express excess deaths as a function of GMT with population scaled in millions: their damage function takes the form $(\beta_1 \cdot T + \beta_2 \cdot T^2) \cdot \text{pop}_{\text{millions}}$. Our function uses the baseline mortality count $M_{ct}$. Additionally, we present excess wildfire $PM_{2.5}$ deaths attributable to climate change—shown in Table 3 from Equation (2). Rather than reporting a similar result based on their quadratic damage function, which enters GIVE, Qiu et al. (2026) report *total* excess deaths from wildfire $PM_{2.5}$, as computed using the Qiu et al. (2025) approach. At 3°C, Qiu et al. (2026) report 64,000 total annual deaths, as compared to the 20,142 annual deaths attributable to climate change that we report in Table 3 replicating their results.

[14] Our discounting is identical to Qiu et al.

Table 3: Average Excess Deaths by Warming Level, Comparison with Qiu et al. 2026

| Temperature | n | Ave. Excess Deaths | Ave. Excess Deaths Qiu et al. |
|---|---|---|---|
| 1.5 | 1,534 | 11,327 | 1,128 |
| 2 | 1,835 | 16,404 | 7,172 |
| 3 | 1,359 | 20,977 | 20,142 |
| 4 | 936 | 32,152 | 37,611 |
| 5 | 426 | 35,976 | 60,897 |
| 6 | 200 | 62,854 | 115,667 |

U.S. annual excess deaths attributable to wildfire $PM_{2.5}$ averaged across all year-iteration pairs where GMT change is $\pm 0.10$°C of the reported warming level.

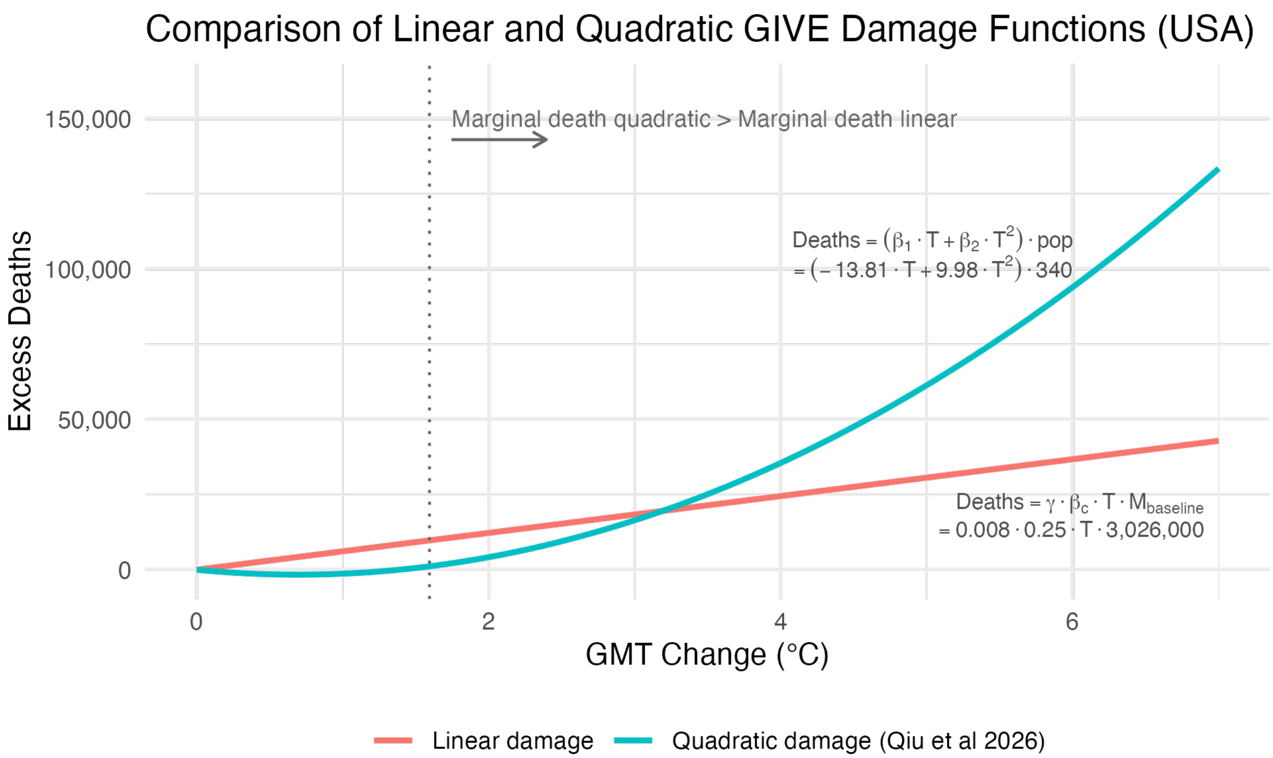


Figure 8: Excess wildfire $PM_{2.5}$ mortality in the USA as a function of GMT change, comparing a linear damage function (this paper) and the Qiu et al. (2026) quadratic function. The dotted vertical line marks the GMT change at which the marginal excess death of the quadratic function exceeds that of the linear (1.59°C). Both functions are evaluated at a fixed U.S. population of 340 million and baseline death rate of 8.9 per 1,000. The $\beta$s for the Qiu et al. (2026) quadratic are representative—averaging over 10,000 values—not the exact parameters entering GIVE. The linear $\beta_c$ is our estimated coefficient for the USA.

# 10 Appendix: Sensitivity analysis

This section interrogates the sensitivity of our results to excluding any one of the EMS informing our analysis. We estimate $\beta_c$ without observations from each of our 5 models. Table 4 shows regression summary statistics: the percentage of countries with positive $\beta_c$ and p-values $<$ 0.05, the minimum, 25th percentile, median, 75th percentile, and maximum. Table 4 provides summary statistics while Table 5 shows $\beta_c$ and p-value results for the top nine most populous countries—which have a larger impact on SC-$CO_2$ results—and the global average. Leaving out CESM: i) most reduces the median $\beta_c$ and the number of significant $\beta_c$ estimates (Table 4); and ii) yields lower $\beta_c$ for populous countries that highly influence the SC-$CO_2$ (Table 5). Accordingly, relative to the main specification, excluding CESM yields the greatest reduction in estimated SC-$CO_2$, lowering the global estimate from \$50.04 to \$25.91 and the USA estimate from \$5.86 to \$0.52 (Table 5). In both cases, although reduced, fire PM2.5 mortality remains the third-largest damage category.

Table 4: Summary statistics of country-level $\beta_c$ estimates across leave-one-model-out sensitivity specifications.

| Model | % $\beta_c > 0$ | % $p < .05$ | Min | 25th | Median | 75th | Max |
|---|---|---|---|---|---|---|---|
| All models | 93 | 65 | -0.41 | 0.08 | 0.20 | 0.40 | 5.23 |
| ex CESM | 87 | 49 | -0.54 | 0.03 | 0.09 | 0.19 | 8.54 |
| ex CLASSIC | 97 | 69 | -0.15 | 0.11 | 0.28 | 0.52 | 5.34 |
| ex JULES | 93 | 63 | -0.46 | 0.08 | 0.22 | 0.43 | 5.42 |
| ex SSiB4 | 85 | 64 | -0.90 | 0.06 | 0.15 | 0.31 | 2.30 |
| ex Zhao | 91 | 65 | -0.47 | 0.08 | 0.22 | 0.46 | 5.90 |

Table 5: $\beta_c$ estimates and standard errors across leave-one-model-out sensitivity specifications for largest countries.

| | All models | | ex CESM | | ex CLASSIC | | ex JULES | | ex SSiB4 | | ex Zhao | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Country | $\beta$ | SE | $\beta$ | SE | $\beta$ | SE | $\beta$ | SE | $\beta$ | SE | $\beta$ | SE |
| global | 0.237 | 0.047 | 0.105 | 0.051 | 0.295 | 0.043 | 0.262 | 0.058 | 0.218 | 0.056 | 0.253 | 0.051 |
| CHN | 0.128 | 0.070 | -0.039 | 0.083 | 0.139 | 0.084 | 0.155 | 0.086 | 0.213 | 0.057 | 0.106 | 0.076 |
| IND | 0.163 | 0.052 | 0.076 | 0.061 | 0.202 | 0.060 | 0.171 | 0.066 | 0.152 | 0.048 | 0.177 | 0.057 |
| USA | 0.253 | 0.170 | 0.018 | 0.234 | 0.286 | 0.215 | 0.284 | 0.216 | 0.304 | 0.070 | 0.279 | 0.186 |
| IDN | 0.138 | 0.040 | 0.217 | 0.047 | 0.118 | 0.039 | 0.146 | 0.049 | 0.144 | 0.049 | 0.094 | 0.040 |
| PAK | 0.250 | 0.143 | 0.354 | 0.207 | 0.275 | 0.181 | 0.276 | 0.181 | 0.107 | 0.018 | 0.280 | 0.157 |
| NGA | 0.384 | 0.492 | 0.588 | 0.719 | 0.814 | 0.323 | 0.351 | 0.626 | -0.177 | 0.524 | 0.428 | 0.541 |
| BRA | 0.200 | 0.081 | 0.132 | 0.099 | 0.234 | 0.085 | 0.221 | 0.098 | 0.166 | 0.086 | 0.219 | 0.088 |
| RUS | 1.313 | 0.326 | 0.119 | 0.021 | 1.490 | 0.404 | 1.493 | 0.403 | 1.483 | 0.405 | 1.501 | 0.350 |
| MEX | 0.155 | 0.047 | 0.099 | 0.054 | 0.181 | 0.052 | 0.163 | 0.059 | 0.146 | 0.048 | 0.167 | 0.051 |

Table 6: Fire PM2.5 SC-CO2 Sensitivity Test ($/tCO2)

| Region | Beta | Mean | 10th | 90th | Spread |
|---|---|---|---|---|---|
| Global | All models | 50.04 | 28.05 | 83.00 | 54.95 |
| Global | Exclude CESM | 25.91 | 10.15 | 47.78 | 37.63 |
| Global | Exclude Zhao | 54.93 | 30.50 | 91.49 | 60.99 |
| Global | Exclude CLASSIC | 65.70 | 33.35 | 108.31 | 74.96 |
| Global | Exclude JULES | 54.31 | 29.41 | 92.05 | 62.64 |
| Global | Exclude SSiB4 | 39.46 | 19.28 | 66.36 | 47.08 |
| USA | All models | 5.86 | 0.71 | 13.24 | 12.53 |
| USA | Exclude CESM | 0.52 | -7.05 | 7.44 | 14.49 |
| USA | Exclude Zhao | 6.46 | 0.79 | 14.58 | 13.79 |
| USA | Exclude CLASSIC | 6.66 | 0.42 | 15.92 | 15.50 |
| USA | Exclude JULES | 6.59 | 0.38 | 15.84 | 15.46 |
| USA | Exclude SSiB4 | 6.94 | 3.23 | 12.81 | 9.58 |

Mean, 10th percentile, and 90th percentile of Fire PM2.5 SC-CO2 across Monte Carlo trials, by region and beta specification.

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